INTRODUCTION

This instruction outlines specific procedures for installation and assembly of a General Electric Model 550 Generator. This document is a guide for use by experienced and qualified staff when performing the work described herein.

WORKSCOPE

The workscope covered in this instruction describes basic field procedures for installing the mechanical, electrical, and instrumentation components of the Generator System as defined in the following Equipment Summary Tables. It also includes specific instructions for the following.

Preliminary Work

- Reviewing Erection Documentation
- Preparing the Work Area

Installing the Equipment

- Installing the Generator Field
- Assembling the End Shields
- Installing the Bearing
- Installing the Collector Housing
- Installing the Lower Frame Extension

Final Installation Inspection

- Verifying Correct Component Installation
- Documenting, Recording and Reporting
- Construction Conformity File

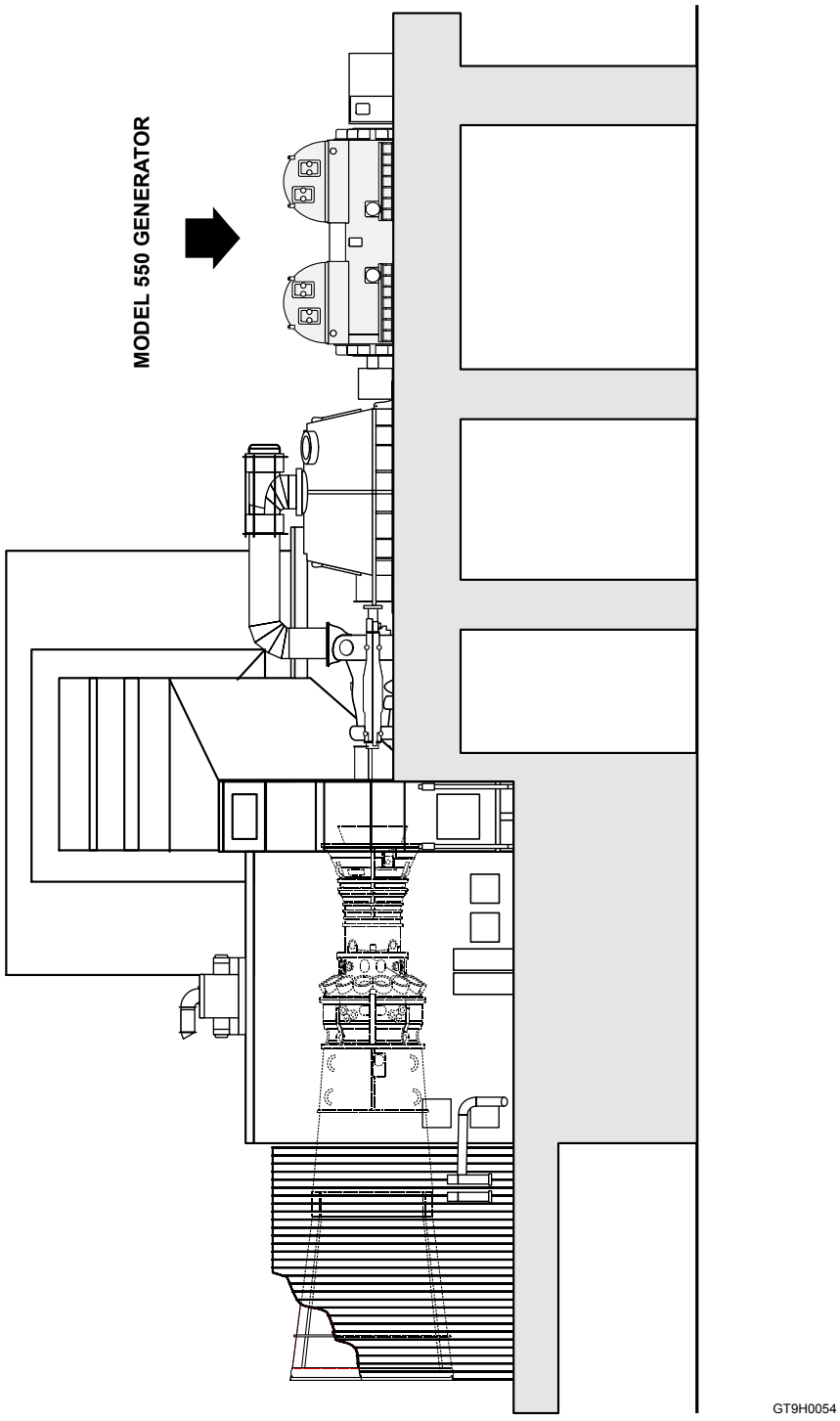


Figure 1. Mechanical Outline – Elevation View (Typical Layout)

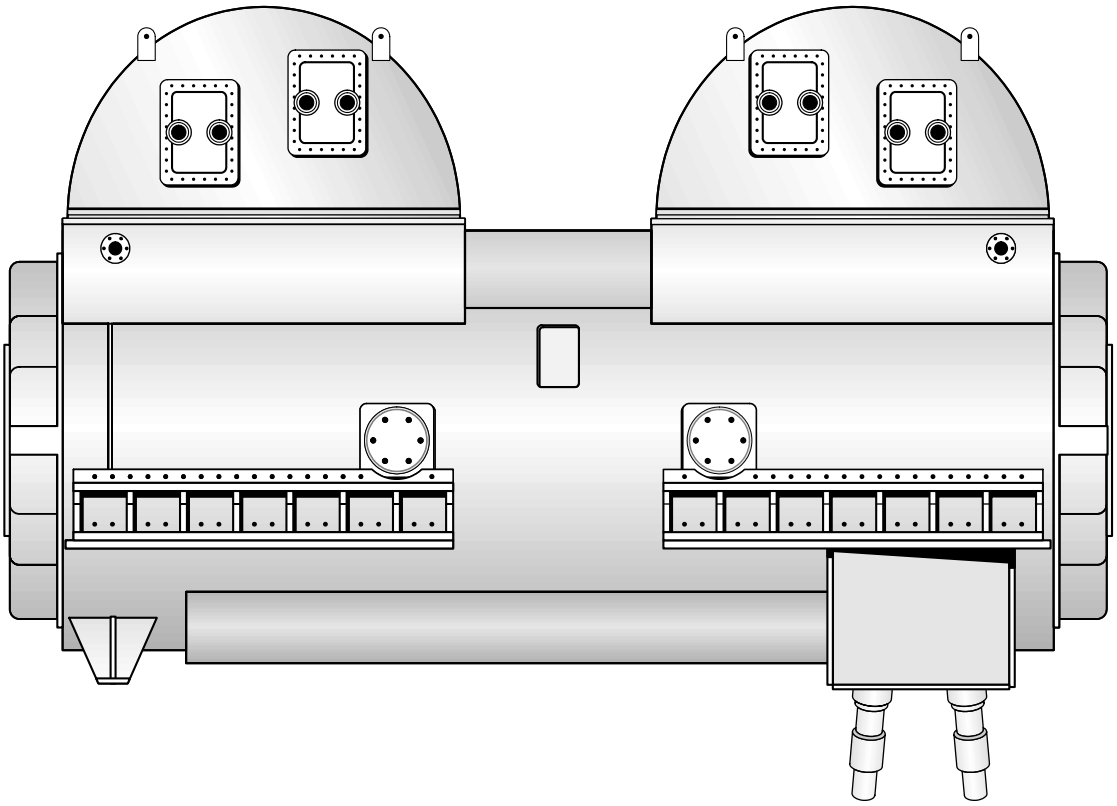


Figure 2. Model 550 Generator with Lower Frame Extension

The following table summarizes the major components or material groups of the Generator System for each unit.

EQUIPMENT SUMMARY TABLE – GENERATOR SYSTEM

MPL Item	Description	Qty.
D4A1	Generator Field	1
B4CB	End Shield - Collector End	1
B4DB	End Shield - Turbine End	1
	Hydrogen Seal Assy. - Collector End	1
	Hydrogen Seal Assy. - Turbine End	1
A3PA	Cooler Clamp Hardware	4
B1H0	Dome Assembly to Stator Frame	1
B1J0	Dome Assembly to Stator Frame	1
B2U0	B2U - Air End Terminal	3
B2V0	B2V - Air End Terminal	3
B3C0	T-BOX HVB Lead Assembly	1
B4C1	Hardware	1
B4D1	Hardware	1
B5A1	Cooler Flanges and Hardware	1
B5B0	Cooler Covers	1
B6A0	RTD and TC Board Connection	1
B6B0	GTG-1 and GTG-3A RTD	1
B6H0	Shaft Monitor Brush Assembly	1
B7E0	Signature Strip	1
B7F0	Data Plate	1
B7J0	Installation Spare Parts	1
B8H0	Assembly Device List	1
D5C0	Lifting Cables	2
E3A0	Base Assembly Brush Holder	1
E3H0	Brush (With Pig Tail)	1

EQUIPMENT SUMMARY TABLE – GENERATOR SYSTEM (Cont'd)

MPL Item	Description	Qty.
E5D0	Filter	1
E7C0	Installation Parts Excitation	1
F1J0	Stator Winding Cooling	1

DESIGN AND ERECTION PHILOSOPHY

Equipment Design Philosophy

The 550 Generator is a hydrogen gas-cooled unit with a separate liquid cooling system for the armature windings. The casing and shields at each end are of welded gas-tight construction. The two-pole rotating generator field is supported in elliptical journal bearings mounted in the end shields. A pair of babbitted rings in each end shield are supplied with oil at a pressure slightly above internal gas pressure to provide a seal for the hydrogen gas. Heat is removed from the field by the circulation of hydrogen gas through the stator and rotary field. The hydrogen gas is cooled by passing through horizontal coolers located in two domes on top of the stator frame. The main power leads are brought out of the casing through six porcelain high voltage bushings.

Erection Philosophy

Throughout erection of the generator, a high level of cleanliness must be maintained to ensure that no dirt or foreign matter is introduced into the generator, and that no tools are left inside the core.

Due to the physical size and weight of the generator, the component parts are shipped separately and erected on site. This philosophy takes into consideration the constraints imposed by the weights and physical size of the component parts of the generator.

The lower frame extension contains factory-assembled high voltage bushings. In order to stay within shipping limitations, the lower frame extension is shipped separately from the generator stator. After insertion of the generator field is completed on site, the lower frame extension is raised into position, bolted to the underside of the generator stator, and seal welded.

The generator stator is lifted and placed on the foundation using special heavy lifting equipment. The rotating field is then lifted and handled using the turbine building overhead crane. After setting the stator, the rotating field is inserted into the stator and the end shields are erected. One electrical connection to each winding lead is then made inside the lower frame extension using bolted flexible connectors. Near completion of mechanical assembly, the end shield piping is assembled.

At this point the brush rigging is installed. Finally, the stator frame lagging is erected and fit between the collector house and the steam turbine lagging.

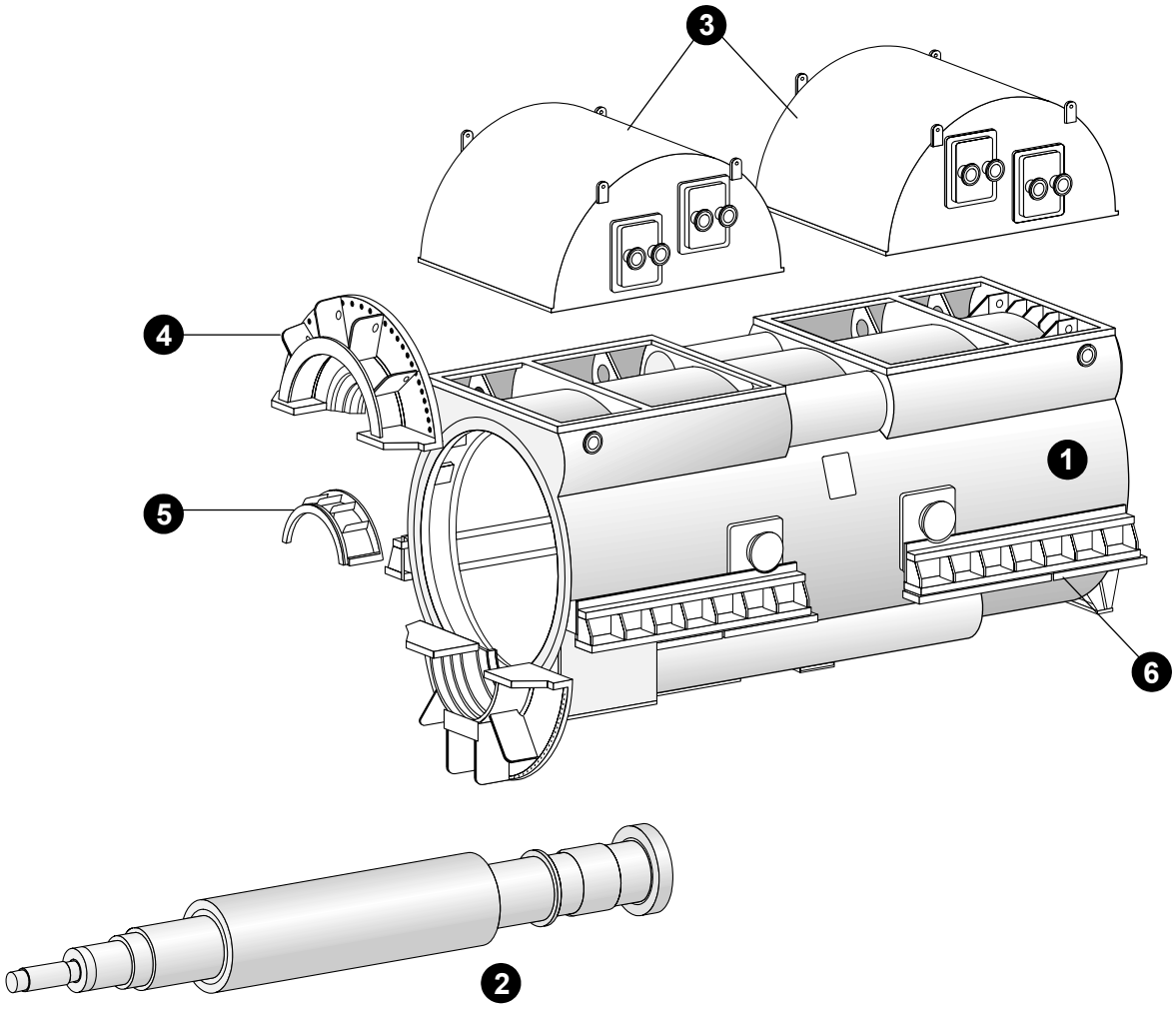
Note: Use only turbine oil to lubricate generator bolting. **Do not use** metallic thread lubricating compounds of any kind.

REFERENCE DOCUMENTS

The following table lists all required drawing and related Material Parts Lists (MPL) documents for installation of the equipment covered in this instruction. These drawings are to be thoroughly reviewed prior to commencing with any of the activities described in this instruction.

MPL Item	Description
A1C	Drawing List, Supplemental - Mechanical Outline
A1CA	Suggested Side Thrust Support
A1CB	Axial Foot Alignment Device
A1CC	End Shield Guide Bracket
A1CD	Stator Core Locking Bolts
A1CE	Weld Cooler Dome to Frame
A1CH	Hydrogen Cooler Outline
A1CK	Generator/Exciter Lagging Support Arrangement
A1CL	Bolt Installation
A1CM	Core Monitor Location
A1C0	Mechanical Outline Support Drawing List
A1D	Electrical Connection Outline
A1D1	Electrical/Mechanical Outline
B6A	RTD and TC Board Connections

MPL Item	Description
B6D	Piping, Generator Connections
B7A0	Gasket Locations
B7GD	Gib Key Plate
B7G	Arrangement Grout Pads - Foundation Plates
B8B	Generator Weights and Volumes
B8HA	Rotor Sling Assembly
B8HB	Seal Assembly Device - Collector End
B8HC	Seal Assembly Device - Turbine End
B8H0	Sealant Gun Assembly
D5A0	Generator Field Clearance
D5B	Field Assembly and Disassembly Notes
D5BB	Installation of Tight Gap Baffle
D5BA	Field Assembly and Disassembly
GEK 103772A	Generator Installation
GEK 7633E	Generator Maintenance
GEK 95110	End Shield Assembly and Disassembly
GEK 75510C	Generator Oil Deflector Assembly
GEK 75520C	Shaft Seal Assembly
GEK 46097D	Journal Bearings
GEI 74489C	Collector, Brush Rigging, Brush Magazine and Brushes
GEK 7618C	Brush Magazine Bus Ring Installation Instructions

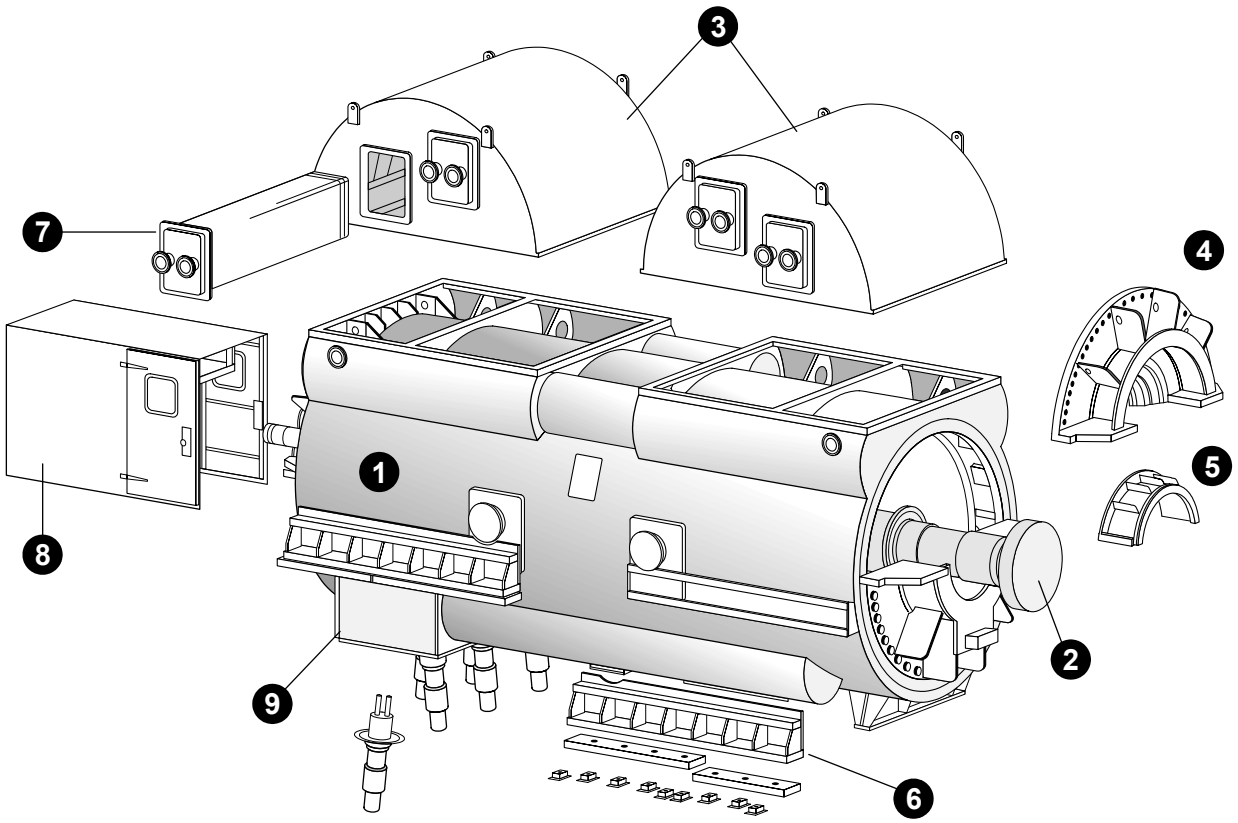


LEGEND

1	Generator Stator	4	End Shield
2	Generator Field	5	Bearing Cap
3	Hydrogen Cooler	6	Generator Foot

GN000065

Figure 3. Generator – Collector End (Exploded View)



LEGEND

1 Generator Stator
2 Generator Field
3 Hydrogen Cooler
4 End Shield
5 Bearing Cap
6 Generator Foot
7 Cooler Bundle
8 Collector Housing
9 Lower Frame Extension

GN000064

Figure 4. Generator – Turbine End (Exploded View)

HEALTH AND SAFETY REGULATIONS IN THE U.K.

The Health and Safety Regulations for the UK **must** be observed during installation of this equipment. Copies of these regulations will be supplied on site by General Electric's consultant (Bucknall Austin). The section covering this article is "Construction - Health, Safety and Welfare Regulations, 1996."

RECEIVING, HANDLING AND STORING EQUIPMENT

The equipment and materials defined in this instruction can normally be stored in an outdoor laydown area with suitable cribbing to elevate the equipment above ground and with adequate weather protection provided (i.e., tarps). Other specific storage or preservation instructions from the Owner or the GE Technical Advisor shall also apply if applicable. All material received shall be stored and arranged to accommodate the correct order of installation. Refer to the detailed erection procedures referenced herein, and to GEZ 5691H, Protection of Steam Turbine and Generator Components During Shipment and Startup.

Inventory of Equipment

The Installer is to perform a detailed inventory of all material against the shipping packing lists provided. In the event a discrepancy, a Discrepancy Report is to be issued. All discrepancies are to be reported immediately to the GE Technical Advisor for verification and assistance in initiating corrective action and filing claims.

All shipping damage or shortages are to be reported immediately to the GE Technical Advisor for verification and assistance in initiating corrective action and filing claims.

Preliminary Work

Prior to commencing with any erection activities, the following preliminary work activities must be completed.

Review of Erection Documentation

1. Review all erection documentation as referenced in this instruction.

Preparation of Work Area

1. Verify that the turbine building overhead crane has been load tested and is operational.

2. Verify that the work area is clear of all debris, equipment, and material that may cause injury to personnel, interference with installation efforts, or damage to the equipment being installed.

3. Verify that all actions referenced above are complete.

EQUIPMENT ERECTION SEQUENCE

Throughout the field assembly, refer to MPL Item D5B, Field Assembly and Disassembly - Notes, and to D5BA, Field Assembly and Disassembly. In the following sections, all Item Nos. mentioned in parentheses ( ) refer to these two drawings. See Figures 5 and 6.

The recommended sequence of erecting the generator is as follows.

1. Preliminary placement of the Lower Frame Extension.
2. Set the Generator Stator (see EM 5486).
3. Remove the temporary End Covers.
4. Install the Inner and Outer End Shields.
5. Install the Generator Field.
6. Align the Generator Field Coupling (per EM 5490).
7. Install the Cooler Bundles into the Stator Dome Coolers.
8. Install the Stator Dome Coolers.
9. Install the Collector Enclosure.
10. Erect the Lower Frame Extension.

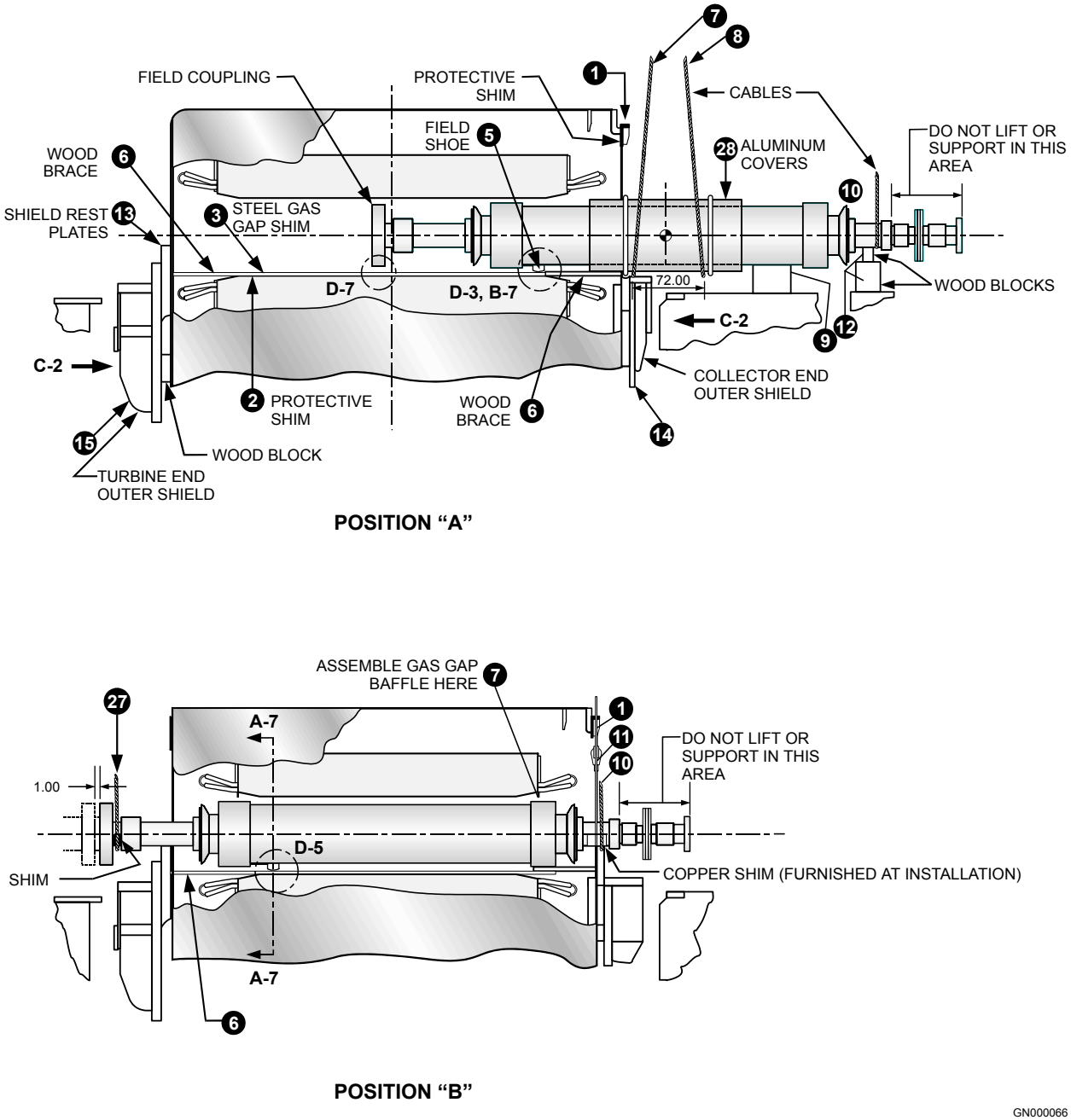


Figure 5. Model 550 Generator – Cross Section View

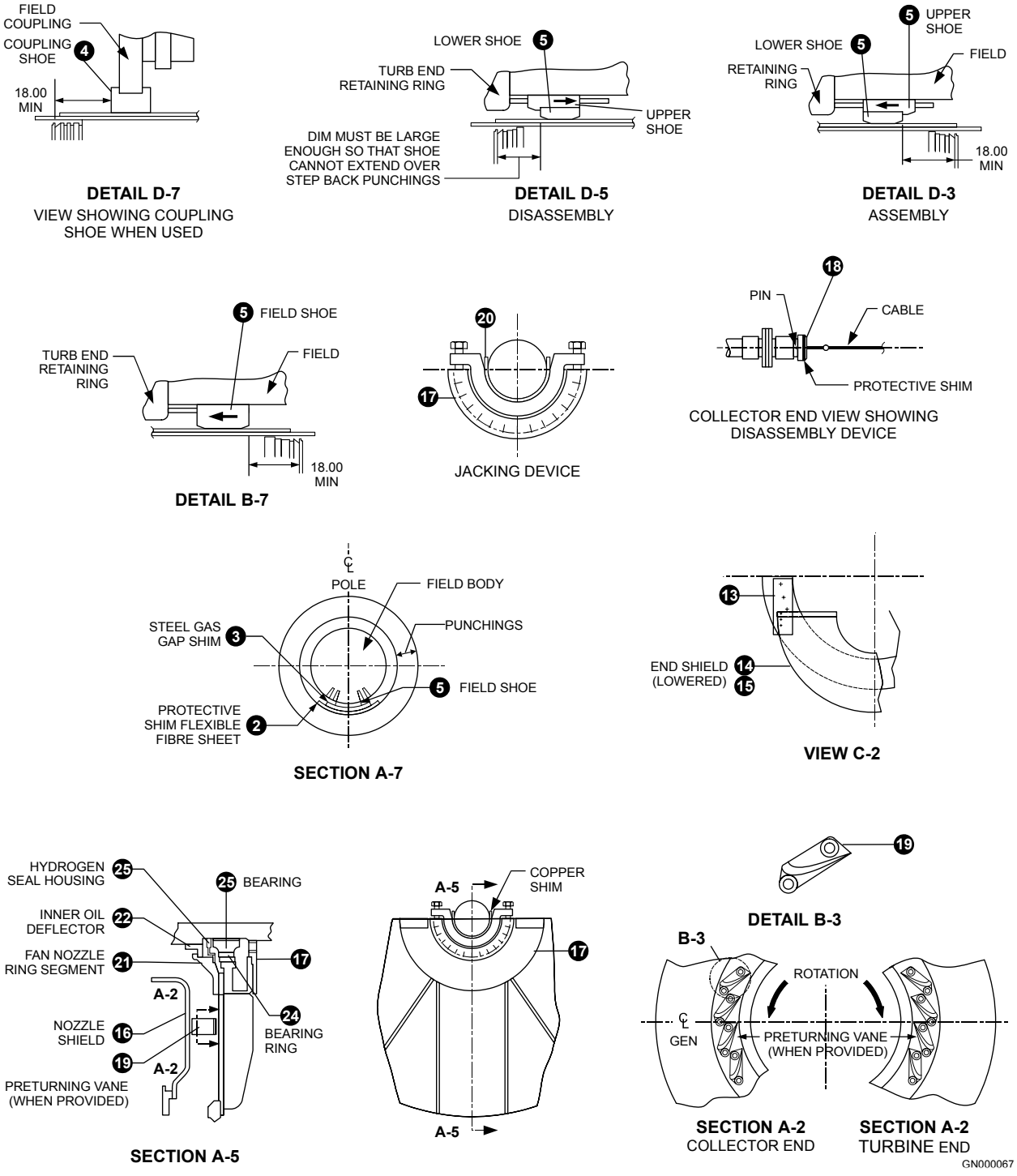


Figure 6. Model 550 Generator Details – Cross Section View

Preliminary Setting of the Lower Frame Extension

The lower frame extension must be positioned in the generator pit prior to setting the generator stator.

Set the lower frame extension on cribbing in the bottom of the generator pit.

Setting the Generator Stator

Setting the generator stator is detailed in EM 5486, MS9001H D-10 Steam Turbine and 550 Generator Centerline Equipment Rigging.

Removing the Temporary Covers

The generator is shipped with temporary steel covers over the ends of the generator stator. These shipping covers must be removed prior to insertion of the generator field.

1. Remove the temporary shipping covers.

2. Clean the Stator to End Shield mating surfaces.

Installing the Inner and Outer End Shields

1. Tools for installing the Generator outer and inner end shields and field into the stator are found in MPL Item B8HO, Assembly Device List.

2. The generator end shields are shipped as complete assemblies with bearings, oil deflectors, and hydrogen seals installed. It will be necessary to partially disassemble these parts and to reassemble them on the generator stator after the field is inserted. Clean and mark all parts and hardware for re-assembly.

3. Protect the stator frame mounting surface using a protective shim.

4. Install the lower end shields at the turbine and collector ends in the lowered position using the lowering brackets (13).

5. Assemble the supporting bracket (1) of the field supporting sling to the collector end of the stator frame.

Installing the Generator Field

Note: Prior to inserting the generator field into the stator, the generator RTDs and TCs should be checked for continuity and function. The generator field should not be installed until all problems with RTDs and TCs are addressed and resolved.

1. Insert the flexible fiber protective shim (2) into the air gap. Grease the inner surface of the steel air gap shim (3) lightly with D6A2C1 (paraffin wax), and place it on the protective shim (2).

2. Ensure that the steel air gap shim (3) is placed with the pulling holes outside the stator core.

3. Verify that all cables are removed from the air gap shim prior to inserting the generator field.

4. Insert a wooden brace (6) between the air gap shim and the end shield so that the shim does not move during insertion of the generator field.

5. With the aluminum covers (28) in place at the middle of the field, and with hitches (7, 8) around the generator field body so that the vertical centerline through a field pole will be at the bottom of the field (as shown in section A-7), enter the field into the core (Position A).

WARNING: Extreme care must be taken when inserting the field into the stator. It is critical that the rotor coupling and the retaining rings do not contact the stator core punchings, as damage to these critical components may occur.

6. Raise the field and insert the Textolite coupling shoe (4). Place the weight of the field on a wooden block (9). The generator field is now supported by the coupling shoe and suitable wooden blocks under the collector end journal. Remove the rigging cables and the protective aluminum cover from the generator field.

7. Rig a suitably sized chainfall and cable between a secure fixed point axially in line with the generator coupling. This chainfall will be used to slide the generator field into the stator while it is resting on the coupling shoe or lower body shoe.

8. Using the cable (10), raise the collector end of the generator field (using a copper shim to protect the journal).

9. Using the axial chainfall installed previously, slide the generator field on the coupling shoe so that the retaining ring is in the correct position for installation of the generator field shoe. Refer to MPL D5B, Section D3 or D5. The house crane will periodically need to be moved toward the generator following the field. (Failure to move the crane can result in stator damage.)

10. With the collector end sufficiently raised, insert the field shoe with the dowels against the retaining ring. Lower the generator field at the collector end to transfer the load from the coupling shoe to the field shoe.

11. Slide the generator field on the shoe until approximately 1 inch (25.4 mm) of space is attained between the generator coupling and LP Steam Rotor flange.

    Note: Extreme care must be used not to nick or damage the coupling face.

12. Assemble the field supporting sling (11) to the supporting bracket (1) at the generator collector end. Support the collector end of the field in the sling and center the field using a protective copper shim to protect the journal.

Installing the Generator End Shields

1. Inspect the end shield flange face and rabbet fit for high spots and burrs. Remove as necessary.

2. Raise the turbine end lower end shield (15) and install per P12A-AL-0223, Outer End Shield Assembly and Sealing Procedures. Clean the bolting surface and apply a light coat of turbine oil on the vertical joint and horizontal joints. Ensure that turbine oil does not get into pumping grooves. Assemble the end shield in position on the stator frame. (The end shield has been assembled, aligned and doweled at the factory.) End shield installation is extremely critical. Carefully follow instructions in P12A-AL-0223. Ensure that the dowels are installed in their original locations as indicated by the stamping on each dowel. Adjustable bushing bolts may be used to position the end shields and ease dowel installation. Torque bolts in sequence per P12A-AL-0223. Torque values for steel generator joint bolts are given in the following table.

Steel Generator Joint Bolts Torque Values

Nominal Diameter (in.)	Threads per Inch	Dry Torque		Lubed Torque	
		ft/lb	Nm	ft/lb	Nm
1/2	13	70	95	50	68
5/8	11	140	190	100	136
3/4	10	250	339	180	244
7/8	9	400	542	300	407
1	8	600	813	450	610
1 1/8	7	750	1017	550	746
1 1/4	7	1100	1491	800	1085
1 3/8	6	1400	1898	1000	1356
1 1/2	6	1900	2576	1400	1898
1 3/4	5	2100	2847	1600	2169
2	4 1/2	3200	4339	2400	3254

Assemble the jacking device (17) on the turbine end lower-half end shield. Fasten the bottom ring to the outer oil deflector bolt circle using hex head cap screws and plain washers. Rig and lift the bottom ring into position using two 1 1/4 x 7 inch eye bolts. Ensure that all cap screws (13) and washers are used to secure the bottom ring to the end shield. Lubricate cap screws before installation and torque to 180 ft./lbs. (244 nm). Clean jack bolts and lubricate with extreme pressure lubricant or approved anti-seize compound before installing in lifting yoke. Rig and lift lifting yoke into position on bottom ring using two 3/4 x 10 inch eye bolts. Before lifting the field with the lifting device, place the copper shim under the generator shaft to protect its surface finish.

3. Raise the turbine end of the generator field enough to unload the field shoe by tightening the jack bolts on the lifting device (seal assembly device). A hitch from shaft to station crane may be used to reduce the force required on the jacking bolts. Raise the collector end of the field with the field supporting sling and remove the field shoe (5), air gap shim (3), and protective shim (2) through the collector end of the stator frame. Center both ends of the field in the stator air gap.

4. Raise the collector end end shield (15) and assemble in position on the stator frame (as performed on the turbine end in the preceding step).

5. Assemble the jacking device (17) on the collector end lower-half end shield. Fasten the bottom ring to the outer oil deflector bolt circle using hex head cap screws and plain washers. Rig and lift the bottom ring into position using two 1 1/4 x 7 inch eye bolts. Ensure that all cap screws (13) and washers are used to secure the bottom ring to the end shield. Lubricate the cap screws before installation and torque to 180 ft./lbs. (244 nm). Clean the jack and lubricate with extreme pressure lubricant or approved anti-seize compound before installing in the lifting yoke. Rig and lift the lifting yoke into position on the bottom ring using two 3/4 x 10 inch eye bolts. Before lifting the field with the lifting device, place the copper shim (20) under the generator shaft to protect its surface finish.

6. Rig up and roll in the lower half of the turbine end ring and bearing. Use two 1 1/2 x 6 inch shoulder eye bolts to lift the bearing half. Lubricate the ball fit and journal surface with oil (type 600 SSU @100F) before rolling the bearing into the bearing ring. Ensure that the bearing is correctly oriented according to the rotation arrow marked on the side of the bearing. Roll the bearing in slowly and carefully, maintaining control by using the two 1 1/2 x 6 inch shoulder eye bolts screwed into the outside diameter at the 40 degree position.

7. Install the lower-half collector end bearing (this is an insulated bearing) using the same procedure as for the turbine end bearing. Ensure that the insulation is not damaged during assembly.

8. Lower the rotor into the bearings and make bearing tilt/twist checks and adjustment.

   At this time, preliminary tilt and twist checks can be made using feeler gauges. Check between journal and bearing working at four at the horizontal joint. Refer to GEK 46097D for tilt/twist limits and calculations.

   The bearings must be in place to obtain an accurate rotor position for setting the inner oil deflectors. After the inner oil deflectors are set, the bearings must be removed to gain access for installation of the hydrogen seals. The bearings are then reinstalled. Final tilt/twist checks are made using lead wire as described in GEK 46097D.

If alignment is found to be not satisfactory, mount a dial indicator on a surface independent of the bearing liner. Place the dial needle to read bearing movement while making adjustments. Adjust by lifting the rotor approximately .025 inch (0.64 mm) from bearing. With weight of rotor off the bearing, tap bearing on horizontal joint using a brass bar or mallet. Place a protective shim on the joint before tapping the bearing. Set rotor weight back in bearing and recheck twist. Repeat as required to obtain proper alignment.

To check the tilt alignment, the upper bearing must be installed. Ensure that joint bolts are tightened so that a .0015 inch feeler will not enter. Perform feeler check between the journal and bearing at locations shown in GEK 46097D.

If alignment is found to be not satisfactory, mount a dial indicator on a surface independent of bearing liner. Place the dial needle to read bearing movement while making adjustments. With weight of rotor off the bearing, tap inboard or outboard sides of the upper bearing liner using a brass bar or mallet. Place a protective shim on the surface before tapping the bearing. Mount a second dial indicator to monitor for twist movement of the bearing while repositioning to fix the tilt. Set rotor weight back in bearing and recheck twist. Repeat as required to obtain proper alignment.

Note: If adjustments are made during the tilt alignment, the twist alignment must be rechecked.

9. With the bearings properly aligned, check the Air Gap Clearance between the stator core and the rotor body at the first fully packed stator section.

10. Lift up the rotor with the jacking device to remove the bearings at the collector and generator ends.

11. Install the lower half of the turbine end gas gap baffle per MPL D5BB. Record the clearances on the Record Data Sheet.

12. Assemble the inner end (nozzle) shields. Hold the lower half against the stator frame section plate when rolling under the field to prevent damage from the pretuning vanes. Leave bolts loose on both upper and lower halves until the horizontal joint plates are installed.

13. Assemble preturning vanes (19) to collector and turbine end lower half end shields, using hex head bolts and lock washers to fasten the vanes. Torque bolts to 150 ft/lbs (203 Nm). One preturning vane on the collector end must be assembled just below the horizontal joint on the left-hand side when viewed from the collector end of the generator (on the turbine end this will be the right-hand side viewed from the collector end). Place remaining 12 around the periphery of lower half end shield.

14. Assemble turbine end lower half inner oil deflector. Clean and roll in lower inner oil deflector (22). Lift using 3/4 x 10 inch eye bolts. Lightly coat the vertical joint gasket with Titeseal and assemble to end shield using the supplied bolts. Adjust deflector clearances to .012 to .015 inch (.035 to .038 mm) on the bottom even on the sides. Secure bolts with lockwire.

15. Clean and assemble upper half oil deflectors on turbine end. Adjust deflectors to achieve drawing clearances.

16. Clean and roll in collector end lower inside oil deflector (22) using 3/4 x 10 inch eye bolts for lifting. Lightly coat the vertical joint gasket with Titeseal and assemble to end shield using bolts with insulated washers and flat washers. Adjust deflector to achieve clearances of .012 to .015 inch (.035 to .038 mm) on the bottom and even on the sides. Use extreme care to avoid breaking insulated hardware by overtightening. Secure bolts with lockwire. Megger check oil deflector half after lockwiring. Minimum megger reading must be 10 Megohm.

17. Clean and assemble upper half oil deflectors on turbine end. Adjust deflectors to achieve drawing clearances.

18. Clean and assemble upper half oil deflectors collector end. Adjust deflectors to achieve drawing clearances. Check inboard oil deflector with a 500 volt megger.

19. Assemble the turbine end hydrogen seal casing and seals. Refer to Shaft Seal Installation and Maintenance Instruction GEK 75520C. Install the seal assembly device if it has been removed. Place indicators on the side and top of rotor. Using shaft jacking device, elevate the rotor very carefully to maintain rotor in center of the end shield. Rotor should not require more than a .010 inch (.25 mm) lift to allow bearing removal (rollout). Roll out bearing lower half. Lubricate the seals and shaft with 600 SSU @100F oil. Roll in lower half of hydrogen seal casing and vertical joint gasket. Lower shaft back to zero position on the indicator. Position the hydrogen seal casing and install bolts. Tighten bolts enough to hold weight of seal casing, and install upper half seal casing and seals. Install the vertical face dowels, and the horizontal joint bolts between upper and lower seal casing halves. Tighten

horizontal joint bolts in sequence from I.D. to O.D. alternating from side to side. See Table 1 of GEK 75520C for torque values. Tighten the vertical joint on the hydrogen seal casing. Remove access cover on upper half hydrogen seal casing and secure both upper and lower halves of hydrogen seals together. Reassemble access covers. Install lockwire on vertical and horizontal joint bolting per GEK-75520C.

Note: Use only turbine oil to lubricate generator bolting. **Do not use** metallic thread lubricating compounds of any kind.

20. Assemble the collector end hydrogen seal casing and seals. Refer to Shaft Seal Installation and Maintenance Instruction GEK 75520C. Install the seal assembly device if it has been removed. Place indicators on the side and top of rotor. Elevate rotor very carefully to maintain rotor in center of end shield. Rotor should not require more than a .010 inch lift to allow bearing removal (rollout). Roll out bearing lower half. Lubricate seals and shaft with 600 SSU @100F oil. Roll in lower half of hydrogen seal casing. Lower shaft back to zero position on the indicator. Position the hydrogen seal casing and install bolts. Tighten bolts enough to hold weight of seal casing. Install upper half seal casing and seals. Install the vertical face dowels to position the seal casing. Install horizontal joint bolts between upper and lower seal casing halves. Tighten horizontal joint bolts in a sequence from I.D. to O.D. alternating from side to side. See Table 1 of GEK 75520C for torque values. Be careful to avoid overtightening the insulated hardware. Tighten the vertical joint on the hydrogen seal casing. Remove access covers on upper half hydrogen seal casing and secure both upper and lower halves of hydrogen seals together. Reassemble access covers. Install lockwire on vertical and horizontal joint bolting per GEK 75520C. After completing installation of collector end hydrogen seal assembly, check resistance across the insulation with a 500 volt megger. Resistance should be no less than 100 Megohms.

21. Clean turbine end bearing cavity and roll in lower bearing ring. Verify that the oil feed holes line up. Clean, lubricate and roll in the turbine end lower bearing half. Using lead wire to check clearances, recheck and adjust bearing twist and tilt clearances as outlined in GEK 46097D. Record final tilt and twist readings.

22. Remove the seal assembly device (rotor jacking device) from the turbine end. Assemble outboard oil deflector on the turbine end. Adjust lower half oil deflector to achieve 1/3 clearance on the bottom and even clearance side to side. Clearance on the bottom should be .012 to .016 inch, clearance on the sides should be approximately .022 inch. Install the upper half oil deflector using the gasket provided and secure the horizontal joint bolts. Measure and record deflector clearances. Remove upper half oil deflector.

23. Clean, inspect and assemble all upper bearing components on the turbine end. Verify that the upper bearing is clean and free of dings, nicks, or scratches. Lubricate and install upper half of the bearing. Flip over the upper bearing cap if required to install the bearing ring. Clean and assemble the bearing ring in the bearing cap. Even the bearing ring horizontal joint and secure bearing to cap by bolting. Rig bearing cap. Clean bearing cap joints and remove any burrs, nicks, or high spots from horizontal and vertical joints. Install bearing cap on end shield using a light coat of light weight Titeseal compound.

24. Assemble upper half oil deflector. Use the gasket provided with the vertical joint. Install horizontal joint dowels bolts. Tighten horizontal joint first, then tighten vertical joint, alternating from side to side working toward the centerline.

25. Clean collector end bearing cavity. Roll in lower bearing ring. Verify that the oil feed holes line up. Clean, lubricate and roll in the turbine end lower bearing half. Recheck and adjust bearing twist and tilt checks (as outlined in GEK 46097D) using lead wire to check clearances. Record final tilt and twist readings.

26. Remove seal assembly device (rotor jacking device) from collector end. Assemble outboard oil deflector on collector end. Adjust lower half oil deflector to achieve 1/3 clearance on the bottom and even clearance side to side. Clearance on the bottom should be .012 to .016 inch, and the sides should be approximately .022 inch. Install upper half oil deflector using gasket provided and secure horizontal joint bolts. Measure and record deflector clearances. Then remove upper half oil deflector.

27. Clean, inspect and assemble all upper bearing components on collector end. Be careful not to damage bearing ring insulation. Verify upper bearing is clean and free of dings, nicks, or scratches. Lubricate and install upper bearing. Flip over the upper bearing cap if required to install the bearing ring. Clean and assemble the bearing ring in the bearing cap. Even the bearing ring horizontal joint and secure bearing to cap by bolting. Megger check bearing ring insulation with a 500 volt megger. Rig up bearing cap. Clean bearing cap joints and remove any burrs, nicks, or high spots from horizontal and vertical joints. Install bearing cap on end shield using a light coat of light weight Titeseal compound.

28. Assemble upper half oil deflector. Install horizontal joint dowels bolts. Tighten horizontal joint first, then tighten the vertical joint, alternating from side to side working toward the centerline.

29. Using the sealant gun assembly, pump Dow Corning Q4-2805 compound into the sealant channels on both end shields per instruction P12A-AL-0223.

Aligning the Generator

1. The next operation would normally be a flush of the Lube Oil System piping and components, thus making lube oil available for rotation of the generator shaft during alignment.

2. Alignment of this unit is accomplished using a method that does not require rotating the generator rotor.

3. Refer to EM 5490, MS9001H Single Shaft Centerline Field Alignment.

4. The guide brackets for the outer end shields will be installed as part of the centerline field alignment.

Installing the H2 Cooler Domes

1. The cooler domes will be shipped separately from the stator frame and will have to be field assembled.

2. If the hydrogen surface coolers are already installed in the cooler domes, proceed to Step 5.

3. Remove the shipping covers from the cooler domes and ensure that all the joints are clean and free from burrs, rust and scale. Remove all material from masked off ventilation openings before installing the coolers.

4. Install the hydrogen surface coolers per MPL Items A3PA, B1JO and B1HO using the supplied hardware (refer to the list in the Appendix).

Note: To ensure proper sealing against the coolers, the cooler bundle rubber baffles must be properly installed for the direction of hydrogen flow.

5. When the coolers are properly installed, mount the Domes on the stator frame per MPL B1HO and B1JO.

Note: Ensure that the stator frame has been properly cleaned before installation of the Domes.

Note: Verify the correct Dome orientation (see below).

- **B1HO Collector End**
- **B1JO Turbine End**

6. After proper bolting of the cooler domes to the stator frame, the domes must be seal welded to the stator frame per welding instruction P8B-AI-8004.

Installation Procedure for HVB Lead Assembly

1. The following material will be required to properly install the HVB.

Description	Qty
Screw, Cap Hex Head	144
Washer, Plain	144
Lock Plate	72
“O” Ring (Viton)	6
“O” Ring (Viton)	6
Elbow HVB	6
Hose, Flex	6
T-Bolt Clamp	12
Flex Lead	36
Washer, Plain	216
Nut, Self-Locking	108
Screw, Cap Hex Head	36
Screw, Cap Hex Head	72
Insulating Boot, HVB	6
Insulating Boot, LWR Lead	6
A50A493C	72
HV Bushing Vent Tube	6
HVB Plate	6
Lock Plate	12
Bolt, Heavy Hex Head	12

“O” Ring (Viton)	6
Compound Thermostat, Flex.	3
Washer, Plain	12
A15F2A	2
A15F6A2	2

2. Install all the above hardware per MPL Item "T-Box HVB and Lead Assembly and Material List."

3. Ensure that the bolts are torqued to 130 ft/lbs (177 Nm). After 8 hours, re-torque to 130 ft/lbs and finally secure by folding the lockplate corners against the bolt heads.

Installing the Collector Housing and Brush Rigging

During installation of the Collector System, refer to MPL Item A1D1, “Collector System Electrical/Mechanical Outline,” and MPL Item E3F, “Brush Holder and Retainer Assembly.”

1. Before starting installation of the collector housing, ensure that all the piping connections to the end shields have been made up and flushed.

2. Ensure that all foundation checks have been carried out and that shims are in the correct position. Also, verify that the shims are free from dirt, burrs, rust or scale.

3. Complete the following procedure only when all information is available for collector installation.

INSTALLING THE LOWER FRAME EXTENSION

The main power leads from the generator are brought out through the lower frame extension to the electrical conductors that connect the generator high voltage output leads (the tubular leads) to the electrical power output, control, protection and delivery system.

1. Ground the factory installed temporary windings wire to the station ground.

2. Remove the factory installed temporary grounding wires connected to the six (6) stator tubular leads (see Figure 7).

3. Using chainfalls, raise the lower frame extension into position under the generator and align the lower frame extension leads to the six (6) stator tubular leads. Install the generator stator frame to the lower frame extension gasket and bolt the lower frame extension to the underside of the generator frame. See Figure 8.

Note: Care must be taken not to damage the air side terminal leads which contain the unit Current Transformers (CTs) and terminal threads. See Figure 9.

4. Seal weld the lower frame extension flange to the generator stator flange.

5. Remove the access hatches on the sides of the lower frame extension.

6. Install the flexible connectors between the six (6) stator tubular leads and the lower frame extension leads.

7. Torque the connection bolts. After 24 hours, re-torque the connection bolts.

8. Tape the flexible connectors.

9. Re-install the access hatches to the sides of the lower frame extension.

10. After installation of the lower frame extension is complete, temporarily ground the generator windings by re-attaching the grounding wires to the air side terminal threads and to the station ground.

CONSTRUCTION CONFORMITY FILE

1. Following complete erection of the generator, perform a complete system installation inspection to verify any outstanding work that must be completed.

2. Verify that all Check Sheets, Inspection Reports, Data Records and all other applicable documentation has been completed in accordance with the Field Quality Standards.



Figure 7. Stator Tubular Leads

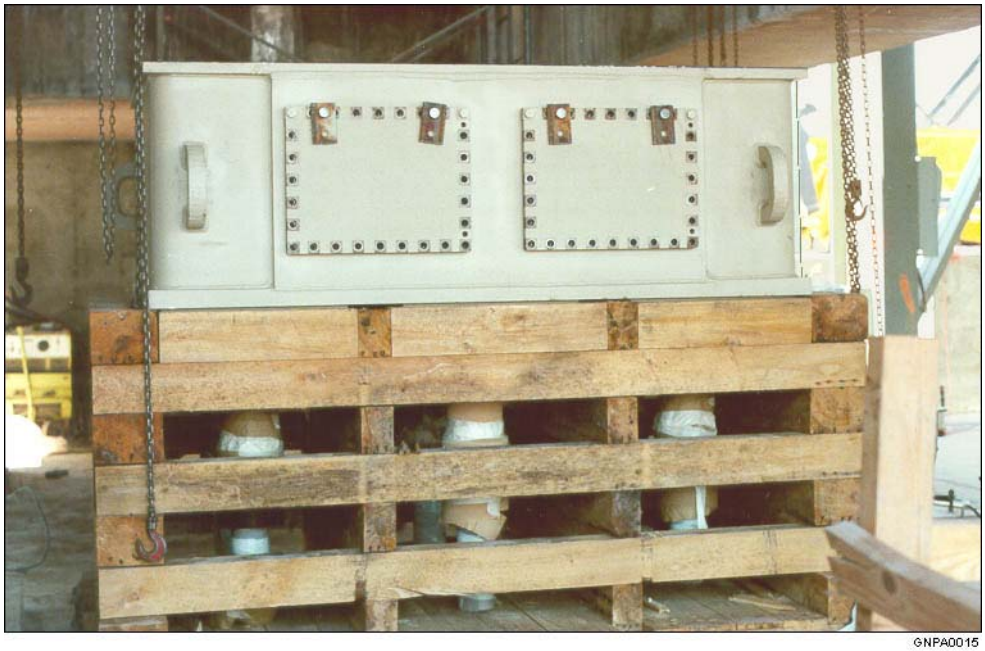


Figure 8. Lower Frame Extension and Current Transformers

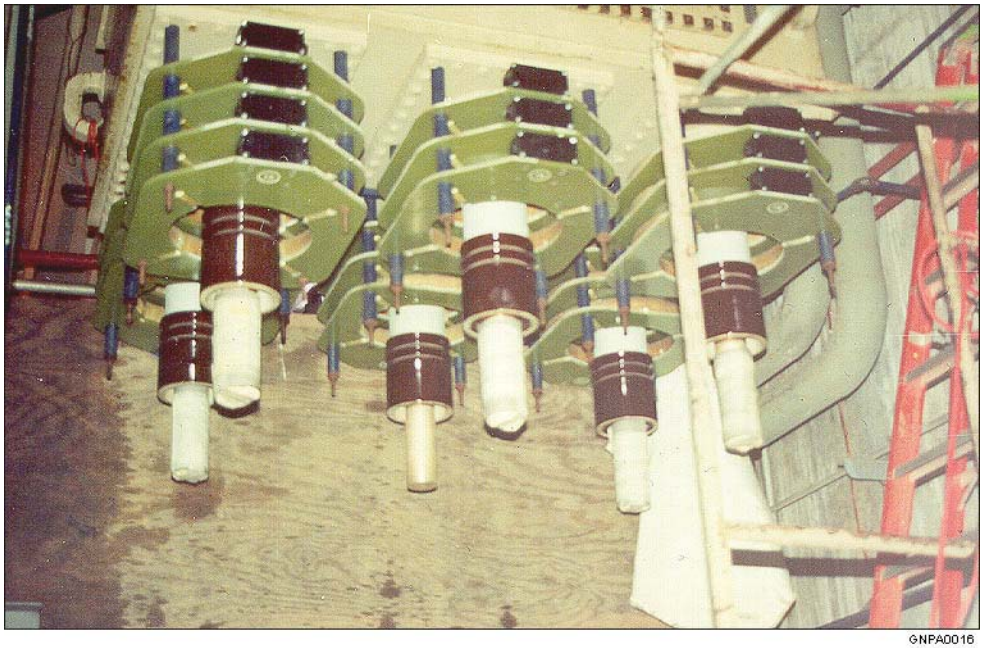


Figure 9. Current Transformers and Airside Terminal Threads

APPENDIX

GENERAL OUTLINE

MPL Item	Description
A1C	Side Thrust Support for Foundation
B8B	Generator Weights and Volumes
D064	Combined Outline, Mechanical Gas, Steam and Generator Combined Outline, Fdn. Interface Gas, Steam and Generator
E7B	Arrangement – Grout Pads and Foundation Plates
GEK 103772	Generator Installation
GEK 33762F	Cleaning and Cleaning Fluids

GENERATOR STATOR INFORMATION

MPL Item	Description
A1CC	Guide Bracket Outer End Shields
A1CD	Locking Bolts, Stator Core Location
A1CJ	Stator Handling
A3PA	Cooler Clamp Assembly
B1H/J	Dome Assembly to Stator Frame
B3F	Nozzle Shield Assembly
B7A	Gasket Locations
P8B-AL-8004	Procedure for Field Welding Cooler Domes
P12A-AL-8003	Stator Foot to Frame Assembly
GEK 7634B	Air Gap Baffle Assembly

GENERATOR FIELD INFORMATION

MPL Item	Description
D5B	Field Assembly and Disassembly
D5BA	Field Assembly and Disassembly Notes
GEK 89504A + ML	Rotor Assembly and Disassembly

GENERATOR END SHIELD

MPL Item	Description
B4C	End Shield ACCU and Shop Fit-Up – Collector End
B4D	End Shield ACCU and Shop Fit-Up – Turbine End
B8H	Seal Assembly Device – Collector End
B8H	Seal Assembly Device – Turbine End
B9J	Package Wiring
GEK 46097D	Journal Bearings
GEK 75510C	Generator Oil Deflector Assembly
GEK 75520C	Shaft Seal Assembly Installation
GEK 95110	End Shield Assembly and Disassembly

GENERATOR COLLECTOR

MPL Item	Description
A1D1	Collector System Electrical/Mechanical
E3B	Base Fabrication Assembly
E3F	Brush Holder and Retainer Assembly
GEI 74489C	Collectors, Brush Rigging, Brush Magazines
GEK 7618C	Brush Magazine Installation Instruction

GENERATOR LAGGING

MPL Item	Description
A1CK	Lagging Support Arrangement

DOCUMENTATION

All inspection items and data records shall be documented and compiled in the unit's Installation and Start-Up Report.